18. Implement "N-Queen Problem" using Backtracking.

```
#include<stdio.h>
int board [20][20], n;
int isSafe[ int r, int c]
{
  int i,j;
  for(i=0;i<c;i++)
     if(board [r][i])
        return 0;
  for (i=r,j=c; i>=0 && j>=0;i--,j--)
     if(board [i][j])
        return 0;
  for (i=r,j=c; i<n && j>=0; i++,j--)
     if(board [i][j])
        return 0;
  return 1;
}
int solve (int c)
 {
    int i;
    if (c==n)
       return 1;
    for (i=0;i<n;i++)
    {
       if (isSafe(i,c))
        {
           board[i][c]=1;
           if (solve(c+1))
              return 1;
           board[i][c]=0;
```

```
    }
  }
  return 0;
}
int main()
{
    int i, j, n;
    printf("Enter number of queens: ");
    scanf("%d", &n);

    if (solve(0))
    {
        printf("\n Solution: \n");
        for(i=0; i<n; i++) {
            for(j=0; j<n; j++)
                printf("%d", board[i][j]);
            printf("\n");
        }
    }
    else {
        printf("No Solution Exists");
    } return 0;
}
```

o/p:

```
Enter number of queens: 4
Solution:
0 0 1 0
1 0 0 0
0 0 0 1
0 1 0 0
```

Partly 27/5/26